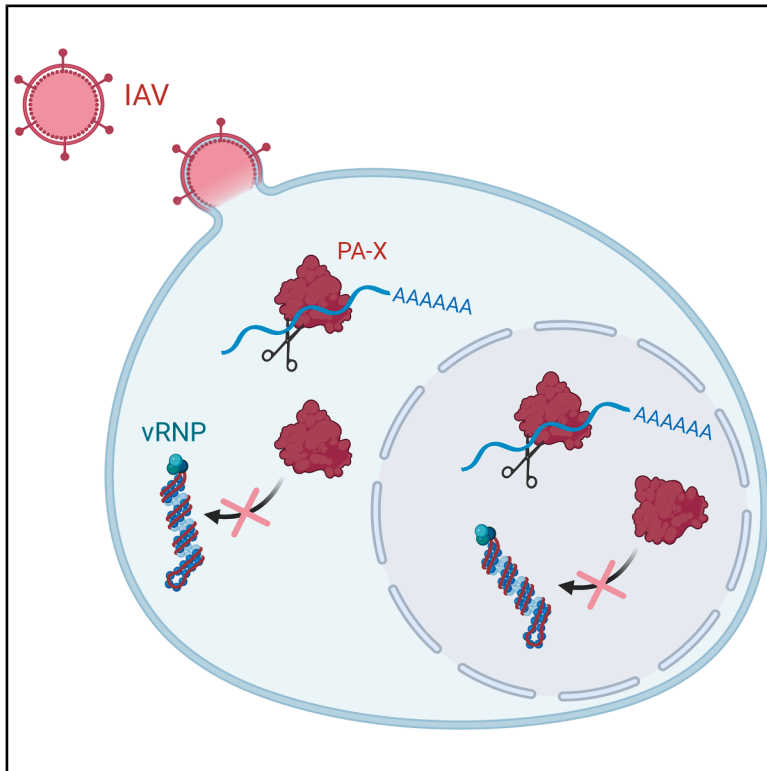


An endonuclease inhibitor unlocks the PA-X shutoff mechanism in influenza A virus

Graphical abstract



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In brief

Oishi et al. successfully purified influenza PA-X to analyze its host shutoff mechanism using an endonuclease inhibitor. We found that PA-X binds RNA via its C-terminal domain to induce degradation, whereas vRNP assembly protects viral RNAs to selectively modulate the cellular environment.

Highlights

- An endonuclease inhibitor enables robust PA-X expression and purification
- PA-X directly binds RNA via its basic-rich C-terminal region
- vRNP assembly protects viral RNAs from PA-X-mediated degradation



Article

An endonuclease inhibitor unlocks the PA-X shutoff mechanism in influenza A virus

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SUMMARY

Influenza A virus protein PA-X mediates host shutoff, yet its molecular mechanism and the role of its C-terminal region remain unclear. When ectopically expressed via plasmids, PA-X cleaves its own mRNAs, leading to remarkably low intracellular expression, which makes its characterization challenging. In this study, we utilize an endonuclease inhibitor to suppress PA-X shutoff activity, enabling its robust expression and purification. Our biochemical analyses reveal that PA-X directly binds RNA via its unique C-terminal basic-rich region and induces mRNA degradation with minimal sequence specificity. Furthermore, we find that while PA-X targets a broad range of RNAs, viral RNAs are protected from its endonuclease activity through their assembly into vRNP complexes. These findings provide critical insights into the PA-X-mediated shutoff strategy, revealing how PA-X modulates the cellular environment while sparing the viral genome to favor viral replication.

INTRODUCTION

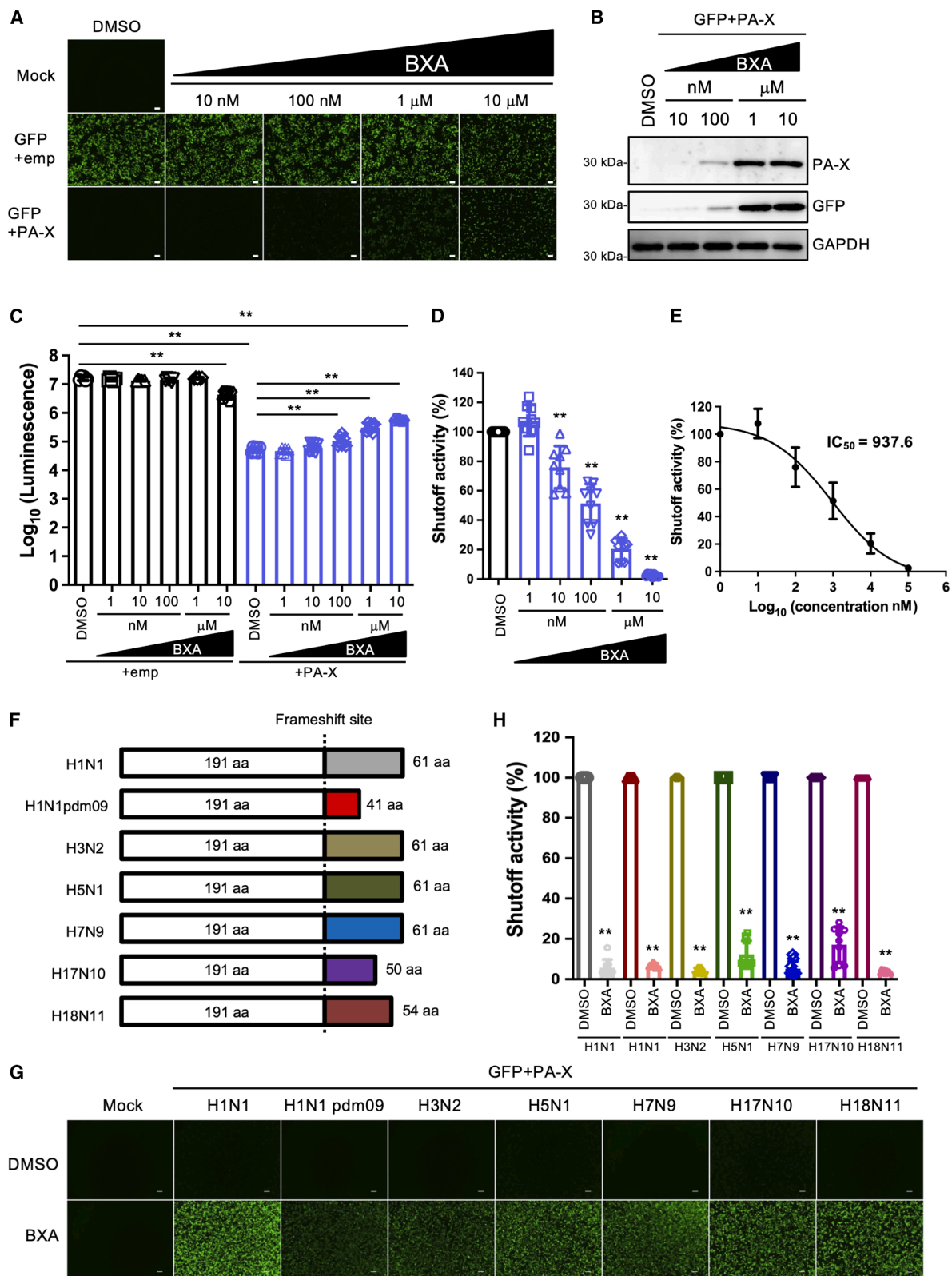
The influenza A virus (IAV) protein PA-X is expressed via a ribosomal frameshift from the PA mRNA.¹ IAV suppresses host gene expression, a process in which PA-X plays a central role through its shutoff activity.¹ By cleaving host mRNAs via its endonuclease activity, PA-X inhibits the production of IFN- β and antiviral antibodies *in vivo*.^{1–7} Although the impact of PA-X on viral replication and pathogenicity remains controversial and varies among strains,^{1–12} recent studies suggest that PA-X facilitates immune evasion by suppressing NF- κ B signaling-associated genes, phenotypic marker maturation, and cytokine secretion.^{4,5,7} Consequently, PA-X is seen as a key modulator of the host response, potentially contributing to virulence and transmission in mammalian hosts.^{1–14}

The N-terminal endonuclease active site, shared by both PA and PA-X, and the unique C-terminal basic amino acids are essential for the shutoff activity of PA-X.^{1,15} Although several studies have been conducted,^{3,15–17} the precise molecular mechanism of PA-X shutoff remains unknown, particularly with respect to the types of nucleic acids targeted, whether PA-X binds to them directly or requires host factors, and how the

C-terminal basic residues facilitate this process. A major challenge in characterizing PA-X *in vitro* is its remarkably low intracellular abundance. This is primarily because PA-X suppresses its own expression from plasmids, while its production from the viral genome is limited by the low frequency of the ribosomal frameshifting event.^{1,2,15,17–20} Attempts to produce large quantities of PA-X in *E. coli* or yeast were hindered by the protein's inherent toxicity. Therefore, mutations have been introduced into the PA-X ORF to reduce its shutoff activity during cultivation.^{18,21,22} Furthermore, N-terminal acetylation by NatB, a posttranslational modification characteristic of eukaryotes, is also important for functional PA-X.¹⁸ Given these constraints, obtaining substantial amounts of wild-type PA-X expressed in mammalian cells is imperative for rigorous *in vitro* characterization.

In this study, we used the endonuclease inhibitor baloxavir acid (BXA) to characterize PA-X. We found that BXA treatment effectively inhibited PA-X shutoff activity, thereby leading to a significant increase in its intracellular expression. Leveraging this phenomenon, we successfully purified substantial quantities of wild-type PA-X from mammalian cells, enabling its detailed *in vitro* characterization. Furthermore, our approach allowed us to elucidate novel properties of PA-X, including its intracellular





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localization, RNA-binding domains, and target specificity. These findings provide key insights into the molecular mechanisms underlying PA-X-mediated shutoff.

RESULTS

The endonuclease inhibitor baloxavir acid (BXA) inhibits PA-X shutoff activity and enhances its expression

Because PA-X cleaves both host mRNAs and its own mRNA, its co-expression with GFP strongly suppressed the GFP signal (Figure S1A). Furthermore, due to this self-targeted cleavage, the level of PA-X remained extremely low despite its overexpression (Figure S1B).^{1,15,18,19} The shutoff activity of PA-X is mediated by its N-terminal endonuclease active site, a domain shared with PA. Consequently, substitution of a key residue within this site, such as E80, abolished this shutoff activity and markedly increased PA-X expression levels (Figures S1A and S1B).^{1–3,15,17–21}

The endonuclease inhibitor baloxavir marboxil (BXM) inhibits viral mRNA synthesis, a process known as cap-snatching, by blocking the endonuclease activity of PAv.²³ Given that PA-X shares this endonuclease activity with PA to cleave mRNAs, we hypothesized that endonuclease inhibitors targeting PA would also inhibit the shutoff activity of PA-X. To test this theory, we examined whether BXA, the active metabolite of BXM,²⁴ inhibits PA-X shutoff activity. When GFP was expressed alone in 293T cells, the GFP signal was clear and stable upon treatment with BXA at concentrations ranging from 10 nM to 1 μ M; however, 10 μ M BXA slightly reduced the GFP signal (Figure 1A). Cell viability analysis demonstrated that BXA treatment up to 1 μ M did not affect viability, whereas 10 μ M BXA was cytotoxic (Figure S1C), suggesting that the reduced GFP expression at the high concentration was due to cell damage. When GFP was co-expressed with PA-X, the GFP signal was strongly suppressed in the absence of BXA. However, this suppression was blocked in a dose-dependent manner in BXA-treated cells (Figure 1A). To confirm these results, protein expression levels were determined by western blotting. Neither GFP nor PA-X was detectable in DMSO-treated cells, whereas the expression of both proteins increased in a dose-dependent manner upon

BXA treatment (Figure 1B). To quantify the inhibition of PA-X shutoff activity by BXA, we used a firefly luciferase reporter assay. While luciferase luminescence was not affected by BXA up to 1 μ M, the addition of 10 μ M BXA reduced luminescence due to its toxicity (Figure 1C; Figure S1C). In cells co-expressing PA-X and luciferase, luminescence was suppressed 100-fold in the DMSO-treated control. Conversely, in BXA-treated cells, this suppression was reversed in a dose-dependent manner (Figure 1C). The shutoff activity of PA-X calculated from the luciferase assay indicates that BXA inhibits PA-X shutoff activity dose-dependently (Figure 1D), with an IC₅₀ of 937.6 nM (Figure 1E). Recently, Case et al. reported that BXA inhibits the shutoff activity of PA-X in a reporter gene co-expression system.²⁵ Our results are consistent with their findings, providing further biochemical evidence that BXA directly interferes with PA-X-mediated shutoff.

Although the nucleotide sequences required for ribosomal frameshifting are highly conserved across all IAV subtypes, and PA-X is thought to be expressed by all subtypes,²⁶ the length of the C-terminal region varies, with PA-X from certain subtypes being C-terminally truncated (Figure 1F). Therefore, we examined whether BXA inhibits the shutoff activity of PA-X proteins derived from diverse IAV subtypes, including H1N1pdm09, H3N2, H5N1, H7N9, H17N10, and H18N11 (Table 1). When each PA-X variant was co-expressed with GFP, the GFP signal was strongly suppressed in DMSO-treated cells but was restored in BXA-treated cells (Figure 1G). Furthermore, BXA treatment increased the expression levels of both GFP and each PA-X variant (Figure S1D). Luciferase assays further demonstrated that BXA treatment reduced the shutoff activity of PA-X from every tested subtype to less than 25% (Figure 1H). While BXA inhibited the shutoff activity of all tested PA-X proteins, the efficiency of signal recovery varied slightly among strains (Figures 1G and 1H), potentially reflecting differences in the baseline activity of each PA-X and its sensitivity to BXA. These data indicate that BXA broadly inhibits the shutoff activities of PA-X across diverse isolates. Collectively, our results show that the endonuclease inhibitor BXA effectively inhibits PA-X shutoff activity, thereby enhancing its intracellular expression.

Figure 1. The endonuclease inhibitor baloxavir acid (BXA) inhibits PA-X shutoff activity and enhances its expression

(A) Representative fluorescent images of 293T cells co-transfected with a GFP expression plasmid and either an empty vector or PA-X expression plasmid. Cells were simultaneously treated with DMSO or the indicated concentration of BXA. GFP signals were observed at 48 h post-transfection. Scale bars indicate 200 μ m. (B) Western blot analysis of GFP and PA-X expression in 293T cells treated with DMSO or BXA. (C) Luciferase activity in cells co-transfected with a firefly luciferase plasmid and either an empty plasmid or a plasmid encoding PA-X. Cells were treated with DMSO or BXA at the time of transfection and incubated for 48 h. The luciferase activities are presented as mean values $\pm$ SD [$n = 9$ ($n = 3$ technical replicates, $n = 3$ biological triplicate)]. ** $p < 0.01$ (one-way ANOVA followed by Tukey's test). (D and E) Inhibition of PA-X shutoff activity by BXA. The shutoff activity of PA-X was calculated by dividing the luminescence value of firefly luciferase co-transfected with an empty plasmid by that co-transfected with the PA-X-expressing plasmid. The shutoff activity of PA-X in DMSO-treated cells was set to 100%. Shutoff activities are presented as mean values $\pm$ SD [$n = 9$ ($n = 3$ technical replicates, $n = 3$ biological triplicate)], and (E) The IC₅₀ was calculated using a non-linear regression analysis with a four-parameter logistic sigmoidal curve. (D)** $p < 0.01$ (one-way ANOVA followed by Dunnett's test). (F) Comparison of PA-X proteins derived from the specific IAV strains used in this study. Schematic representations of the C-terminal region of the PA-X proteins of the strains listed in Table 1 (representative subtype H1N1, H1N1pdm09, H3N2, H5N1, H7N9, H17N10, and H18N11) are shown. (G) GFP signals in 293T cells co-expressing PA-X derived from various IAV subtypes. Cells were treated with DMSO or 1 μ M BXA at the same time as plasmid transfection, and GFP fluorescence was observed at 48 h post-transfection. Scale bars indicate 200 μ m. (H) Comparative shutoff activities of PA-X from the indicated IAV strains (H1N1, H1N1pdm09, H3N2, H5N1, H7N9, H17N10, and H18N11) in the presence of DMSO or BXA. The shutoff activity of each PA-X in DMSO-treated cells was set to 100%. The shutoff activities are presented as mean values $\pm$ SD [$n = 9$ ($n = 3$ technical replicates, $n = 3$ biological triplicate)]. ** $p < 0.01$ according to a two-tailed unpaired Student's t test. (A, B, and G) Images are representative of at least three independent experiments.

Table 1. Influenza A virus strains of each subtype

Subtype	Strain	GenBank
H1N1	A/Puerto Rico/8/34	EF467820.1
H1N1pdm09	A/California/04/2009	MN371611.1
H3N2	A/Panama/2007/99	CY112922.1
H5N1	A/human/Zhejiang/16/2006	EF100817.1
H7N9	A/Shanghai/02/2013	KF918658.1
H17N10	A/Little yellow-shouldered bat/Guatemala/2010	CY103891.1
H18N11	A/Artibeus lituratus/Brazil/2301/2012	MH682202.1

Purification of recombinant PA-X and *in vitro* characterization of its single-stranded nucleic acid specificity

Having demonstrated that BXA increases PA-X expression by inhibiting its shutoff activity (Figure 1B), we sought to leverage this phenomenon to produce sufficient quantities of PA-X protein in mammalian cells for *in vitro* characterization. Expi293F cells were transfected with a plasmid expressing PA-X derived from the A/Puerto Rico/8/34 (H1N1) strain in the presence of 1 μ M BXA, a concentration previously determined to inhibit PA-X shutoff activity effectively without inducing cytotoxicity (Figure 1B; Figure S1C). Following protein purification, the PA-X protein was subjected to SDS-PAGE. CBB staining revealed a distinct single band at approximately 30 kDa (Figure 2A). The identity of the purified protein was further confirmed by western blotting using an anti-PA monoclonal antibody that recognizes the N-terminal region (Figure S2A).²⁸ These results demonstrate that we successfully purified substantial amounts of recombinant wild-type PA-X protein.

To determine whether the purified PA-X protein retained its functionality, we first assessed its endonuclease activity using single-stranded DNA (ssDNA) as a substrate, following previously established protocols for evaluating the PA N-terminal domain.²⁹ Consistent with the properties of the PA N-terminal domain, the endonuclease activity of PA-X was dependent on divalent cations, particularly manganese (Mn^{2+}), with weaker activity observed in the presence of magnesium (Mg^{2+}) (Figure 2B).²⁹ In contrast to its effect on ssDNA, PA-X failed to cleave either linear or circular double-stranded DNA (dsDNA) (Figure S2B). To confirm that the observed ssDNA degradation was mediated specifically by PA-X and not by contaminating host nucleases, we performed the reaction in the presence of inhibitors that specifically target the influenza virus PA endonuclease.^{29,30} The addition of BXA inhibited PA-X-induced ssDNA degradation in a dose-dependent manner; specifically, 1 μ M and 10 μ M BXA partially and completely inhibited PA-X endonuclease activity, respectively (Figure 2C). In contrast, 10 μ M 2,4-dioxo-4-phenylbutanoic acid (DPBA), a conventional influenza endonuclease inhibitor, failed to inhibit ssDNA cleavage by PA-X. This result is consistent with previous findings indicating that BXA is a more potent inhibitor of the PA endonuclease than DPBA.³¹ While both DPBA and BXA act via metal ion chelation,³¹ only BXA had a significant inhibitory effect under these conditions. The lack of inhibition by DPBA may be due to its lower potency or binding affinity for PA-X compared to BXA, consistent with a previous report.³²

A previous study evaluated the *in vitro* endonuclease activity of PA using its purified N terminus 209 amino acids (PA209).²⁹ Since the PA-X-specific C-terminal domain is also required for potent shutoff activity,^{1,3,15} PA-X exhibited significantly higher shutoff activity than PA209 when co-expressed with reporter genes such as GFP or luciferase (Figures S2C and S2D). To determine whether the endonuclease activity measured *in vitro* correlates with the shutoff activity observed in cells, we purified PA209 and assessed its activity using the same methodology (Figure S2E). Although PA209 partially degraded ssDNA in the presence of Mn^{2+} , as previously reported,²⁹ its endonuclease activity was markedly lower than that of PA-X; specifically, a much smaller amount of PA-X was sufficient to cleave ssDNA compared to PA209 (Figure 2D; Figure S2F). These data demonstrate that while PA-X and the PA N-terminal domain share common endonuclease properties due to their identical active sites, their catalytic potencies differ significantly. Furthermore, these results indicate that the endonuclease activity evaluated *in vitro* directly corresponds to the shutoff activity exhibited within cells.

We next assessed the endonuclease activity of purified PA-X toward single-stranded RNA (ssRNA). When incubated with an unstructured ssRNA substrate, PA-X exhibited cleavage only in the presence of Mn^{2+} (Figure 2E). Similar to the results observed with ssDNA, ssRNA cleavage was inhibited by 1–10 μ M BXA, but was unaffected by the same concentrations of DPBA (Figure S2G). In contrast, dsRNA was not cleaved by PA-X regardless of the divalent cations present (Figure S2H). Given that PA-X also failed to cleave dsDNA (Figure S2B), its endonuclease activity appears to specifically target single-stranded nucleic acids. To further investigate the structural specificity of PA-X, we performed an endonuclease assay using various structured RNA substrates, including transfer RNA (tRNA), signal recognition particle (SRP) *Alu* RNA,³³ and pan-handle RNA (ph-RNA) derived from IAV genome sequences (Table 2)³⁴ (Figures 2F–2J). While tRNA and *Alu* RNA are relatively resistant to degradation by PA209 compared to ph-RNA,²⁹ PA-X cleaved these structured RNAs with comparable efficiency (Figures 2F–2H). To directly compare PA-X-mediated cleavage, we analyzed control and PA-X-treated RNA substrates on the same gel. The original full gel image (provided as Figure S2I) encompasses the data shown in Figures 2G and 2H, as well as additional results for non-structured RNA. Although identical amounts of each RNA substrate were used, minor variations in initial band intensities were observed, likely due to sequence-dependent

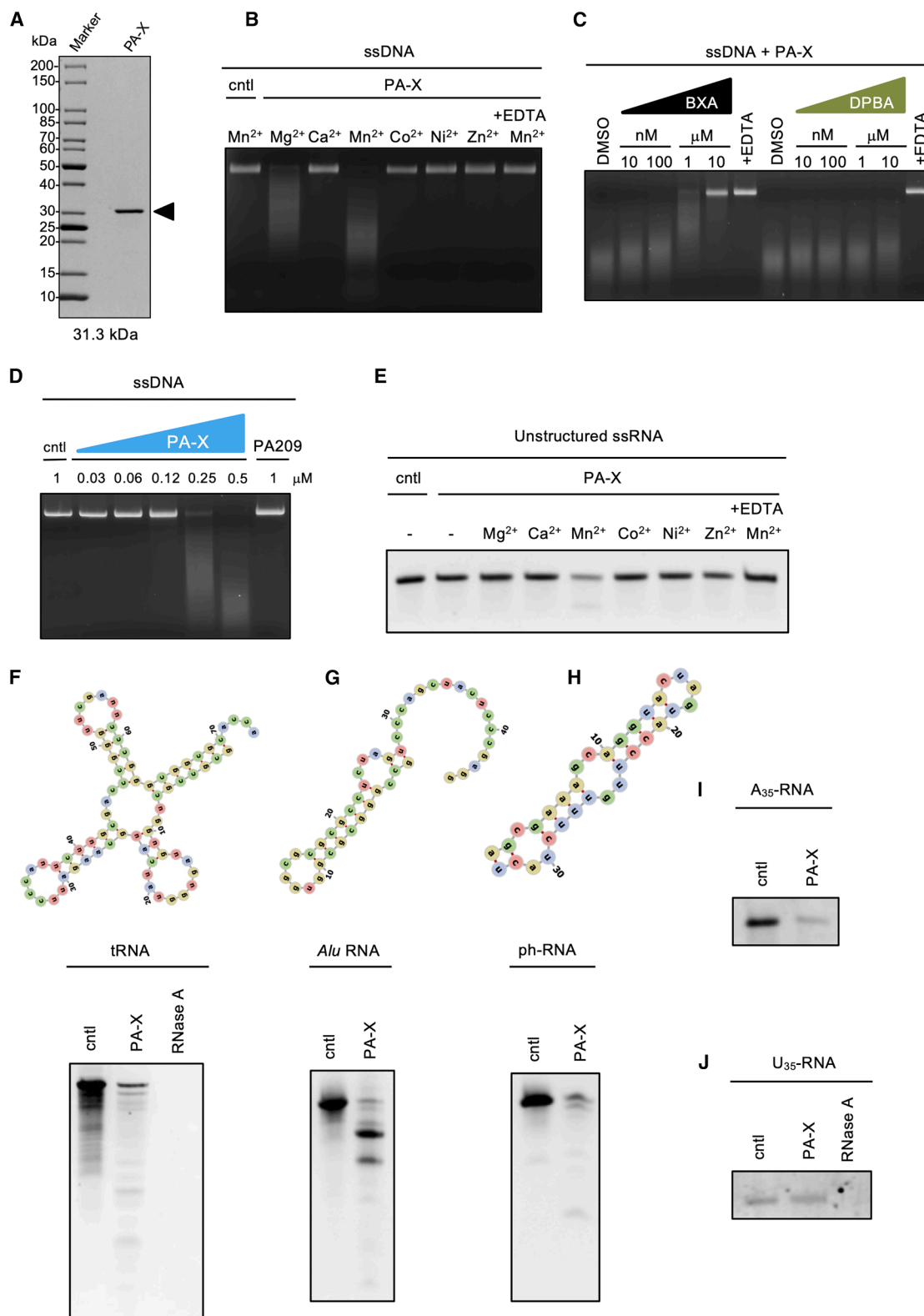


Figure 2. Purification of recombinant PA-X and in vitro characterization of its single-stranded nucleic acid specificity

(A) SDS-PAGE analysis of purified PA-X. PA-X expressed in Expi293F cells was purified by immunoprecipitation and subjected to SDS-PAGE through a 5%–20% gel followed by Coomassie Brilliant Blue (CBB) staining.

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differences in staining efficiency (Figure S2I). Despite these variations, PA-X exhibited comparable cleavage efficiency for both unstructured and structured RNAs, such as ph-RNA and *Alu* RNA. When tRNA was incubated with PA-X, unique band patterns were observed on electrophoresis (Figures 2F–2H), whereas incubation with RNase A resulted in complete degradation (Figure 2F). To explore sequence specificity, we examined the activity of PA-X toward homopolymeric RNA substrates, such as poly A and poly U. PA-X cleaved A₃₅-RNA, which consists of 35 adenine ribonucleotides (Figure 2I). In contrast, U₃₅-RNA was relatively resistant to PA-X-mediated cleavage, despite being efficiently degraded by RNase A (Figure 2J). To objectively evaluate PA-X cleavage efficiency, we quantified the band intensities (Figure S2J). Consistent with the gel images, most RNA substrates were efficiently cut by PA-X, although U₃₅-RNA showed relative resistance (Figures 2G–2I; Figure S2I). While tRNA was clearly degraded, it was modestly resistant in the quantitative analysis (Figure 2F; Figure S2J), likely because the overlapping multiple bands observed in the control lanes and produced by partial PA-X cleavage made it difficult to resolve and quantify the individual fragments.

These data suggest that PA-X cleaves single-stranded nucleic acids with varying efficiencies depending on the ribonucleotide composition, but shows a preference for adenine over uracil.

Viral RNAs resist PA-X-mediated shutoff by forming vRNP complexes

Despite extensive research, the precise targets of PA-X have remained elusive; for instance, whether PA-X selectively cleaves specific host RNAs or degrades viral RNA (vRNA) is still unclear. Many viruses selectively degrade host mRNAs by exploiting structural differences from viral mRNAs, such as the presence or absence of a cap structure or a poly A tail.^{35–37} Other viruses induce global mRNA degradation to fine-tune the viral life cycle.^{38–40} Because PA-X cleaves host mRNAs in infected cells,^{1–7} we synthesized RNAs with or without a cap structure and/or a poly A tail via *in vitro* transcription to clarify whether these modifications influence PA-X activity *in vitro*. When each RNA substrate was incubated with PA-X protein, PA-X induced degradation regardless of a cap structure or a poly A tail *in vitro* (Figure 3A). These data indicate that PA-X targets and cleaves RNA independently of a cap structure or a poly A tail, at least *in vitro*.

To investigate whether PA-X targets vRNAs, we prepared vRNAs and incubated them with purified PA-X. IAV amplified in

MDCK cells was concentrated by ultracentrifugation, and vRNAs were isolated from viral particles (Figure 3B). When the vRNAs were incubated with PA-X, the copy numbers of the NP segment were significantly reduced compared to those incubated with the control protein. Although the addition of BXA led to a slight increase in vRNA copy numbers, the difference did not reach statistical significance (Figure 3C). These results suggest a potential trend where BXA may inhibit PA-X-mediated degradation of vRNAs, although this remains to be validated. We also examined vRNA segments other than that of NP and found that PA-X similarly degraded both HA and NA vRNA and that their degradation was partially inhibited by BXA (Figure S3A). In contrast, RNase A treatment led to the complete degradation of all vRNA segments (Figure 3C; Figure S3A).

In infected cells, vRNAs do not exist in a naked form but are packaged into viral ribonucleoprotein (vRNP) complexes through interactions with the viral polymerase (PB2, PB1, and PA) and NP.^{41–43} To determine whether PA-X can cleave vRNAs within these complexes, we isolate vRNPs from virus particles using glycerol density gradient ultracentrifugation as described elsewhere (Figure S3B).^{44,45} Notably, when vRNPs were incubated with PA-X, the vRNAs within the vRNP structures were resistant to PA-X-mediated cleavage, in stark contrast to the susceptibility observed with naked vRNAs (Figure 3D). Unlike PA-X, RNase A successfully degraded the vRNAs within the vRNPs, as previously reported (Figure 3D).⁴⁵ Similar resistance was observed for the HA and NA vRNA segments within vRNPs (Figure S3C). PA-X cleaved ssDNA in the presence of the vRNP fractions (Figure S3D), ruling out the possibility that components in the RNP fraction, such as glycerol, inhibited the endonuclease activity of PA-X. Collectively, these data indicate that while PA-X cleaves a broad range of RNAs, including mRNAs and naked vRNAs *in vitro*, it does not cleave vRNAs within vRNP structures, suggesting that vRNAs resist PA-X-mediated shutoff in infected cells by forming vRNP complexes.

The basic amino acids in the PA-X-specific C-terminal region function as an RNA-binding motif to facilitate host shutoff

Previously, we reported that six highly conserved basic amino acids in the unique C-terminal region of PA-X are essential for its shutoff activity.¹⁵ However, how these basic residues contribute to PA-X function has remained unclear. To investigate the role of these residues, we constructed a mutant PA-X (PA-X

(B) Divalent cation requirement for PA-X endonuclease activity. 10 nM of ssDNA was incubated with 0.5 μ M purified PA-X or BSA as a control in the presence of 1 mM MgCl₂, CaCl₂, MnCl₂, CoCl₂, NiCl₂, or ZnCl₂ at 37°C for 2 h. Reactions were stopped with loading dye containing 10 mM EDTA and analyzed by 1.0% agarose gel electrophoresis and GelRed staining.

(C) Inhibition of PA-X endonuclease activity by BXA. PA-X-mediated degradation of ssDNA was assessed in the presence of 1 mM MnCl₂ and increasing concentrations of BXA or DPBA (10 nM, 100 nM, 1 μ M, and 10 μ M). DMSO and 20 mM EDTA served as negative and positive controls, respectively.

(D) Comparison of the endonuclease activities of PA-X and PA209. The indicated amounts of PA-X, 1 μ M PA209, or BSA (control) were incubated with ssDNA in the presence of 1 mM MnCl₂ at 37°C for 2 h.

(E) Divalent cation dependence of PA-X RNase activity. 2.5 μ M unstructured ssRNA was incubated with 2.5 μ M PA-X or BSA under the indicated conditions. Reaction products were denatured at 70°C for 10 min and analyzed by 20% urea-PAGE.

(F–J) Substrate specificity of PA-X. Secondary structures of RNA substrates were predicted using MXfold2 (<https://ws.sato-lab.org/mxfold2/>).²⁷ 2.5 μ M tRNA (tRNA), highly structured *Alu* RNA, pan-handle RNA (ph-RNA), A₃₅-RNA, or U₃₅-RNA were incubated with 2.5 μ M PA-X or BSA in 1 mM MnCl₂. 0.25 μ M RNase A was used as a control for tRNA and U₃₅-RNA degradation. Following incubation at 37°C for 2 h, reactions were quenched with loading dye-containing EDTA, denatured, and analyzed by 20% urea-PAGE. RNA sequences are provided in Table 2. (A–J) All gel images are representative of at least three independent experiments.

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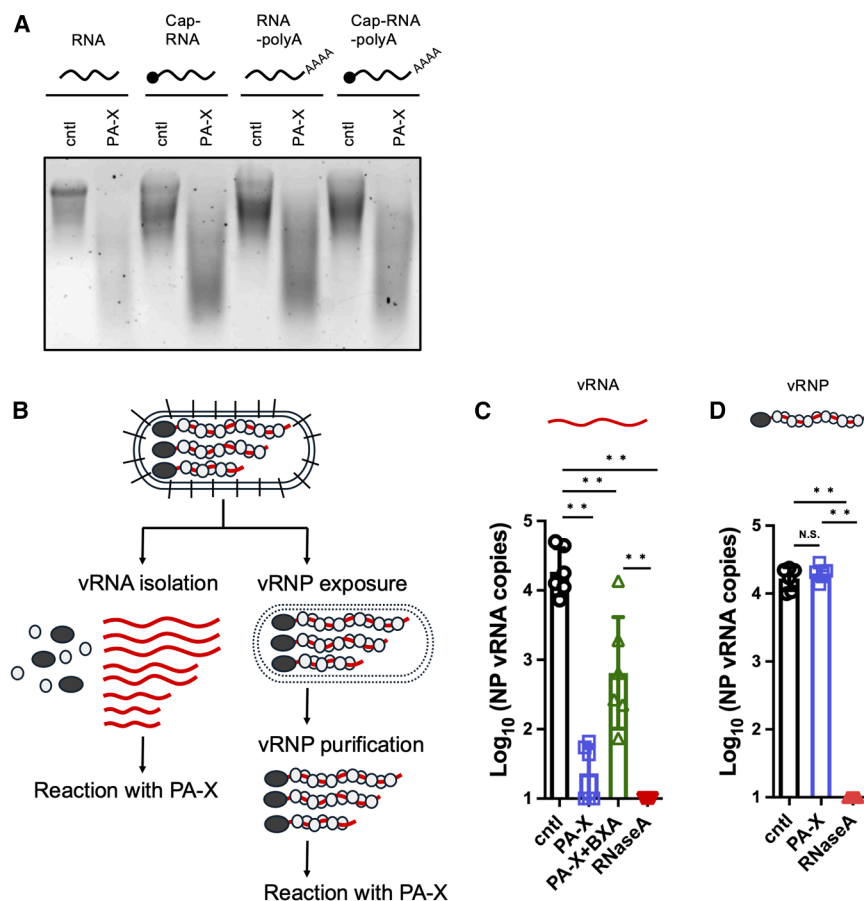


Figure 3. Viral RNAs resist PA-X-mediated shutoff by forming vRNP complexes

(A) Effect of a cap structure or poly A tail on PA-X endonuclease activity. RNA substrates with or without a cap structure and/or a poly A tail were synthesized by *in vitro* transcription. 0.5 μ M of each RNA was incubated with 5.0 μ M PA-X or BSA in the presence of 1 mM MnCl₂. The gel image is representative of at least three independent experiments.

(B) Experimental workflow for assessing PA-X activity against naked viral RNAs (vRNAs) and vRNP complexes. Schematic representation of the preparation of naked vRNAs and intact vRNP complexes isolated from IAV particles.

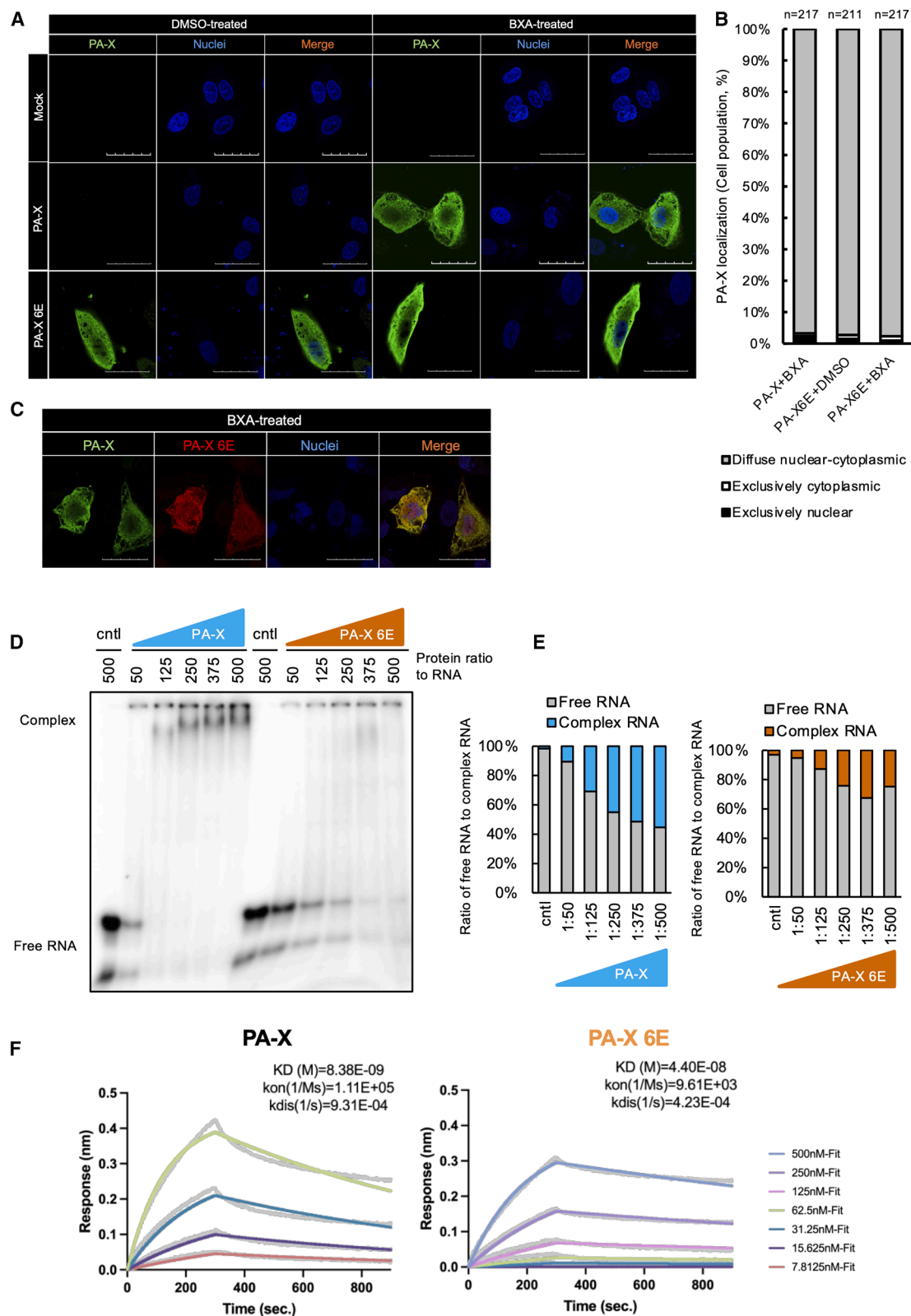
(C and D) Resistance of vRNP complexes to PA-X-mediated cleavage. Naked vRNAs (C) or vRNP complexes (D) were incubated with 1.5 μ M PA-X in the presence of 1 mM MnCl₂ and either DMSO or 1 μ M BXA. In place of PA-X, 1.5 μ M BSA and 0.15 μ M RNase A were used as a control. Following a 2-h incubation at 37°C, total RNA was isolated, and copy numbers of the NP vRNA segment were quantified by RT-qPCR. The RT-qPCR data are presented as mean values $\pm$ SD [$n = 6$ ($n = 2$ technical replicates, $n = 3$ biological triplicate)]. ** $p < 0.01$; N.S., not significant (one-way ANOVA followed by Tukey's test).

RNAs, such as the poly A tail, and that the six basic amino acids mediate its interaction (Figure S5A).

To quantify the interaction between PA-X and RNA, the binding profiles of recombinant wild-type PA-X and PA-X 6E for the highly structured RNA were measured using biolayer interferometry. Wild-type PA-X bound to the highly structured RNA more effectively than PA-X 6E, due to a more efficient association rate (Figure 4E). Binding profiles for A₃₅-RNA revealed a similar trend (Figure S5B). Specifically, wild-type PA-X exhibited a significantly stronger binding affinity for A₃₅-RNA compared to the PA-X 6E mutant, as reflected by their respective binding curves and KD values. These findings were validated by additional biological replicates using independently purified proteins, confirming the robustness and reproducibility of the observed differences in binding kinetics (Table S1). In addition, to examine whether BXA inhibits PA-X activity by

interfering with its RNA-binding capability, we evaluated the PA-X-RNA interaction in the presence of BXA (Figures S5C and S5D). Both the EMSA and BLI analyses demonstrated that PA-X binds to RNA efficiently regardless of BXA, suggesting that BXA inhibits the subsequent endonuclease activity rather than the RNA-binding step.

Taken together, these results indicate that the six basic amino acids in the PA-X-specific region contribute to its shutoff activity by facilitating direct binding to RNAs. Notably, while the PA-X 6E mutant exhibited a significant reduction in shutoff activity and RNA-binding affinity compared to wild-type PA-X, it retained a detectable level of residual activity and substantial binding affinity (Figure 4F; Figures S4A and S4D). This suggests that although the six basic residues are essential for efficient function, other residues within PA-X likely also contribute to its interaction with RNA.



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DISCUSSION

In this study, we conducted a comprehensive *in vitro* characterization of PA-X by producing recombinant PA-X protein in mammalian cells supplemented with the endonuclease inhibitor BXA. Our analysis revealed fundamental properties of PA-X, including its intracellular localization, RNA-binding capacity, and substrate specificity. Based on these data, we propose a molecular mechanism for PA-X shutoff activity: PA-X distributes to both the cytoplasm and the nucleus, where it binds to various RNAs via its C-terminal basic-rich region and cleaves a broad range of ssRNA without strict specificity. While PA-X targets both host and viral mRNA,⁵¹ vRNAs are protected from PA-X-mediated cleavage through their assembly into vRNP structures. Moreover, PA-X limits its own expression levels through an auto-suppressive function, which prevents unintended loss of essential viral transcripts. Additionally, PA-X effectively suppresses host immune responses by targeting the transcripts of essential immune-related genes that are strongly induced upon infection. This transcript degradation likely modulates the cellular environment to create an environment favorable for viral replication by optimizing viral fitness while silencing antiviral defenses.

The clinical use of the polymerase inhibitor BXM has led to the emergence of resistant variants.⁵² The highest level of resistance is associated with the I38T and I38M substitutions in PA, both of which are located in the endonuclease active site shared by PA and PA-X. A recent study showed that these mutations, as well as BXA, reduce the shutoff activity of PA-X, which is consistent with our findings.²⁵ Interestingly, these mutations attenuate PA-X shutoff while preserving viral polymerase activity, suggesting that the effects of amino acid substitutions near the endonuclease active site differ between PA and PA-X.

Virus-induced host shutoff refers to the widespread downregulation of host gene expression and has emerged as an important process in the reallocation of cellular resources for viral replication and evasion of host antiviral immune responses.³⁶ The targeting of host mRNAs by viral nucleases is a common shutoff strategy employed by both DNA and RNA viruses.^{35,53} While some viruses selectively degrade host mRNAs by exploiting

structural differences between host and viral transcripts, or through the temporal and spatial regulation of viral protein expression,³⁷ others employ a global mRNA degradation strategy to fine-tune host and viral mRNA levels, which is essential for their replication cycle.^{38–40} Identifying the specific nucleic acid targets of PA-X is essential to understand its functional role in infected cells. When PA-X is co-expressed with other proteins via plasmids, the expression of both the target proteins and PA-X itself is strongly suppressed, indicating a lack of strict sequence specificity for RNA degradation by PA-X, at least for mRNA.^{2,5,12,15,17,18,20,21} Comprehensive transcriptome analyses comparing wild-type and PA-X-deficient viruses *in vivo* support this concept and suggest that PA-X preferentially suppresses genes involved in innate immune and inflammatory responses.^{1–3,5–7} These findings imply that PA-X targets a broad spectrum of abundant mRNAs without strict specificity, mainly targeting host mRNAs to facilitate host shutoff while retaining the potential to affect viral mRNAs in infected cells.⁵¹ Our *in vitro* analysis demonstrated that PA-X cleaves a wide range of ssRNA, including influenza vRNAs, with the exception of U₃₅-RNA, which consists of 35 consecutive uracils. Consistent with these results, no meaningful binding specificity for unique RNA sequences or structures was observed. However, despite the permissive specificity of PA-X, vRNAs organized into vRNP structures exhibited marked resistance to PA-X-mediated cleavage, unlike naked vRNAs. Given that RNase A can degrade vRNAs within vRNPs,⁴⁵ this resistance appears to be specific to PA-X. We speculate that the PA-X shutoff strategy involves global mRNA degradation, potentially including viral mRNAs, while vRNAs remain sequestered and protected from PA-X activity within vRNP complexes. In this way, PA-X shutoff modulates the cellular environment to favor viral replication without strict substrate specificity. Nevertheless, it should be noted that *in vitro* analyses may not fully reflect the complex interactions mediated by host factors, other viral materials, or spatial regulation within cells.^{54–57} Specifically, although PA-X exhibits the catalytic potential to cleave viral mRNA *in vitro*, this activity does not necessarily imply that these mRNAs are its primary targets during viral infection. Therefore, further investigations will

Figure 4. The basic amino acids in the PA-X-specific C-terminal region function as an RNA-binding motif to facilitate host shutoff

(A) Intracellular localization of wild-type PA-X and the PA-X 6E mutant. Huh-7 cells were transfected with FLAG-tagged PA-X or PA-X 6E and treated with either DMSO or BXA. At 48 h post-transfection, PA-X was detected by immunofluorescence using an anti-FLAG antibody and an Alexa Fluor 488-conjugated secondary antibody. Nuclei were stained with Hoechst 33342. Images were acquired using an FLUOVIEW FV3000 confocal microscope. Scale bars indicate 20 μ m.

(B) Quantification of the subcellular localization of PA-X and PA-X 6E. The intracellular localization patterns of wild-type PA-X and the PA-X 6E mutant in DMSO- or BXA-treated cells were quantified and classified into three categories: exclusively nuclear, exclusively cytoplasmic, or diffuse nuclear-cytoplasmic. Data represent the percentage of cells in each category ($n > 200$ cells were analyzed across three independent experiments).

(C) Colocalization of wild-type PA-X and PA-X 6E. Cells were co-transfected with plasmids expressing FLAG-tagged PA-X and His-tagged PA-X 6E. At 48 h post-transfection, proteins were detected using an anti-FLAG mouse monoclonal antibody and an anti-His rabbit monoclonal antibody, respectively. Localization of PA-X and PA-X 6E was determined by using Alexa Fluor 488-conjugated goat anti-mouse IgG and Alexa Fluor 555-conjugated goat anti-Rabbit IgG as secondary antibodies. Nuclei were stained with Hoechst 33342. Scale bars indicate 20 μ m.

(D) Electrophoresis mobility shift assay (EMSA) of the PA-X-RNA interaction. ³²P-labeled highly structured ssRNA (*A**lu* RNA) was incubated with the indicated amounts of purified PA-X or PA-X 6E at room temperature for 1 h. BSA served as a negative control. Protein-RNA complexes were resolved on a 6% native polyacrylamide gel.

(E) Quantification of the band intensity of EMSA. The ratios of free RNA and complex RNA were quantified using Fiji software.⁵⁰

(F) Kinetic analysis of PA-X-RNA binding by biolayer interferometry (BLI). Binding affinities of wild-type PA-X and PA-X 6E for structured RNA were determined using the Octet RED96e instrument. Gray lines in the panels indicate the association and dissociation of PA-X proteins at concentrations ranging from 62.5 nM to 7.8125 nM for PA-X or 500 nM–7.8125 nM for 6E PA-X (2-fold serial dilutions). Color lines indicate globally fitted sensorgrams used to obtain the association rate (*k*_{on}), the dissociation rate (*k*_{dis}), and the KD values shown at the top of each graph. (Representative data of two technical replicates; see also Table S1). (A, C, and D) All images are representative of at least three independent experiments.

be needed to fully elucidate PA-X function in the context of infection, particularly given the challenges posed by its low expression via ribosomal frameshifting and its inherent self-regulatory properties.

One of the strengths of the current study is the successful expression and purification of recombinant wild-type PA-X using a mammalian cell system. While isopropyl β -D-1thiogalactopyranoside (IPTG)-inducible protein expression in *E. coli* is a standard method for obtaining large quantities of protein, both our group and others have reported that PA-X expression in *E. coli* and yeast frequently drives the emergence of mutations that reduce the shutoff activity of PA-X.^{18,21,22} Expressing PA-X in *E. coli* thus appears to exert selective pressure that favors the emergence of mutations. Such mutations, which typically reduce PA-X shutoff activity to alleviate its inherent toxicity, can be introduced during cultivation. This often results in a protein pool dominated by attenuated mutants rather than the authentic wild-type protein, thereby compromising the reliability of subsequent analyses. While leaky expression can be mitigated to some extent by using specialized *E. coli* strains, the potential for spontaneous mutations remains a concern, necessitating sequence verification of expression plasmids before and after the production process. To overcome these limitations and ensure the integrity of the recombinant protein, we instead employed a transient expression system in mammalian cells to produce wild-type PA-X.

During the production of recombinant PA-X in mammalian cells, BXA was added to the cell culture medium 18 h post-transfection, and no steps were taken to remove BXA during protein expression and immunoprecipitation. Despite that, the purified PA-X retained its functional endonuclease activity. Furthermore, the *in vitro* enzymatic activity of the purified PA209 and PA-X 6E correlated with their shutoff activities observed in expressed cells (Figure 2D; Figures S2 and S4). These results suggest that the addition of BXA during protein production has no effect on subsequent *in vitro* analyses of recombinant PA-X protein. The endonuclease activities of PA and PA-X are dependent on two essential divalent metal ions (Figures 2B and 2E), and BXA coordinates with these catalytic metal ions within the active site to inhibit enzymatic function.^{52,58} We speculate that while BXA effectively inhibits PA-X shutoff activity within the intracellular environment, it is subsequently dissociated from the protein during purification. This dissociation may occur as BXA competes for binding with other metal ions or reagents, such as imidazole during elution, thereby restoring the functional integrity of the recombinant PA-X for downstream applications.

Transient expression of PA-X in the presence of BXA allowed us to identify its intracellular location. Previous studies have suggested a predominantly nuclear localization for PA-X because its target mRNAs are cleaved within the nucleus. To overcome detection challenges, these studies utilized fluorescent protein fusions combined with highly sensitive microscopy or nucleocytoplasmic fractionation.^{16,17} While biochemical fractionation is highly sensitive, it can introduce artifacts; for instance, overexpressed proteins forming insoluble inclusion bodies may cosediment with the nuclear fraction, resulting in false-positive identification of nuclear localization. In our study, when PA-X was expressed and observed in the absence of BXA, the signals appeared predominantly in the nucleus, with faint signals in the

cytoplasm (Figure S4D). However, these signals were too faint to definitively determine the localization of PA-X, given the possibility of imaging artifacts such as nuclear signal spillover. To resolve this issue, PA-X expression levels were increased through BXA treatment. Under these conditions, we demonstrated that PA-X localizes to both the nucleus and the cytoplasm (Figure 4A). Although BXA did not alter the localization of the PA-X 6E mutant (Figure 4A), we cannot entirely exclude the possibility that BXA might influence the distribution of wild-type PA-X. Our results also revealed inconsistencies with previous reports. A previous fractionation-based study suggested that basic residues in the C-terminal region contribute to the nuclear localization of the PA-X of the A/California/04/2009 (H1N1pdm09) strain.¹⁷ In contrast, our immunofluorescence analysis showed that the PA-X 6E mutant derived from the A/Puerto Rico/8/1934 (H1N1) strain retains its nuclear localization (Figure 4A). These differences may reflect both technical and biological factors. Notably, in the previous study, all PA-X constructs carried a K134A mutation to abolish endonuclease activity and enhance protein expression, whereas we used wild-type PA-X. Because even a single amino acid substitution can influence subcellular localization, differences in sequence backgrounds, along with inherent strain-specific traits, likely contributed to the observed discrepancies. These findings suggest that the requirement of specific residues for nuclear entry may vary depending on context, or that redundant nuclear import mechanisms exist. Furthermore, although the localization of PA-X mutants has been used to predict that of wild-type PA-X, these mutants may not accurately reflect the authentic localization due to the unintended effects of the mutations on its localization.^{12,17,21,49} Further studies with more sensitive imaging techniques are required to visualize the localization of wild-type PA-X in naive or infected cells.

A previous study employed 5' RACE-seq to compare RNA fragments from cells infected with wild-type or PA-X-deficient IAV and identified specific fragments enriched in wild-type-infected cells, leading to the suggestion that PA-X preferentially cleaves RNA at GCUG.⁵⁹ However, the interpretation of these results was complicated by high background noise and the potential involvement of other host endonucleases, which were not excluded from the assay. Moreover, PA-X-mediated modulation of host gene expression, including that of endogenous nucleases, may have confounded the analysis. In contrast, our study used a direct *in vitro* approach to examine PA-X activity on various RNA substrates containing GCUG sequences, including unstructured RNAs, tRNAs, and relatively long transcripts. Under these conditions, we observed no significant cleavage sequence specificity by PA-X (Figures 2 and 3A). This discrepancy may stem from methodological differences: while the previous study assessed reporter RNA cleavage within a complex cellular environment via plasmid transfection and RT-PCR,⁵⁹ our analysis was conducted entirely *in vitro*. By using purified recombinant protein, our system eliminated the cellular context and potential confounding factors, allowing for a more direct evaluation of the catalytic properties of PA-X.

In summary, here we demonstrated that the endonuclease inhibitor BXA enhances PA-X expression by inhibiting its shutoff activity. The production of recombinant PA-X in mammalian cells

supplemented with BXA enabled a detailed *in vitro* characterization of PA-X, shedding light on the molecular mechanisms underlying its shutoff function. This study provides a robust methodology for preparing large quantities of PA-X protein using BXA, which will be a valuable tool for future biochemical and structural studies. Although further studies are required to fully elucidate the role of PA-X in viral infection, our findings lay the foundation for understanding how this protein modulates the host-pathogen interface.

Limitations of the study

The use of the endonuclease inhibitor BXA enabled a detailed biological analysis of PA-X; however, several limitations remain. Our *in vitro* characterization revealed that PA-X cleaves RNA without strict sequence specificity, suggesting a broad catalytic potential that likely includes viral mRNA. Nevertheless, these findings do not necessarily imply that such mRNAs are the primary targets during actual viral infection. Further, our *in vitro* system may not fully replicate the potential effects of other host or viral factors that might cooperate with PA-X to modulate its activity or target specificity, nor does it account for the complex spatial regulation within cells.

While BXA treatment allowed us to determine the intracellular localization of PA-X by enhancing its expression levels, the interpretation of these results requires caution. Although BXA did not alter the localization of the PA-X 6E mutant, the possibility that BXA influences the distribution of wild-type PA-X cannot be entirely ruled out. Therefore, further studies using more sensitive imaging techniques are required to visualize the authentic localization of wild-type PA-X during infection.

Additionally, the extremely low expression of PA-X remains a major obstacle to its characterization in infected cells. While it remains unclear whether PA-X undergoes self-suppression during actual infection as observed in transfected cells, the ribosomal frameshifting required for PA-X production is itself an inherently rare event. Thus, even in the absence of self-suppression, PA-X protein levels would remain minimal. Furthermore, while BXA treatment can potentially enhance PA-X expression by blocking its catalytic activity, it also inhibits viral polymerase activity, which consequently leads to a reduction in overall viral protein synthesis.

Taken together, while our *in vitro* results shed light on the molecular mechanisms of PA-X-mediated shutoff, further investigations are necessary to fully understand its functional role and regulation in the context of a viral infection.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Kohei Oishi (oishi.k@jihs.go.jp).

Materials availability

All materials and reagents will be made available upon execution of a material transfer agreement (MTA).

Data and code availability

- **Data availability:** No unique datasets or code were generated or analyzed in this study.

- **Code availability:** This study did not generate original code. Software and algorithms used for data analysis are listed in the [key resources table](#).
- **All other items:** Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

K.O., K.S., S.Y., and Y.K. designed the study. K.O. and K.S. performed the experiments. K.O. and K.S. analyzed the data. K.O. and K.S. prepared the original draft, and S.Y., K.M., H.H., and Y.K. reviewed and edited the manuscript. All authors reviewed and approved the final manuscript.

DECLARATION OF INTERESTS

The authors declare no conflicts of interest.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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REFERENCES

1. Jagger, B.W., Wise, H.M., Kash, J.C., Walters, K.A., Wills, N.M., Xiao, Y.L., Dunfee, R.L., Schwartzman, L.M., Ozinsky, A., Bell, G.L., et al. (2012). An overlapping protein-coding region in influenza A virus segment 3 modulates the host response. *Science* 337, 199–204. <https://doi.org/10.1126/science.1222213>.

2. Chen, X., Kong, M., Ma, C., Zhang, M., Hu, Z., Gu, M., Wang, X., Gao, R., Hu, S., Chen, Y., et al. (2025). The PA-X host shutoff site 100 V exerts a contrary effect on viral fitness of the highly pathogenic H7N9 influenza A virus in mice and chickens. *Virulence* 16, 2445238. <https://doi.org/10.1080/21505594.2024.2445238>.
3. Hayashi, T., MacDonald, L.A., and Takimoto, T. (2015). Influenza A Virus Protein PA-X Contributes to Viral Growth and Suppression of the Host Antiviral and Immune Responses. *J. Virol.* 89, 6442–6452. <https://doi.org/10.1128/jvi.00319-15>.
4. Qin, T., Chen, Y., Huangfu, D., Miao, X., Yin, Y., Yin, Y., Chen, S., Peng, D., and Liu, X. (2023). PA-X protein of H9N2 subtype avian influenza virus suppresses the innate immunity of chicken bone marrow-derived dendritic cells. *Poult. Sci.* 102, 102304. <https://doi.org/10.1016/j.psj.2022.102304>.
5. Hu, J., Kong, M., Cui, Z., Gao, Z., Ma, C., Hu, Z., Jiao, X., and Liu, X. (2020). PA-X protein of H5N1 avian influenza virus inhibits NF-kappaB activity, a potential mechanism for PA-X counteracting the host innate immune responses. *Vet. Microbiol.* 250, 108838. <https://doi.org/10.1016/j.vetmic.2020.108838>.
6. Avanthay, R., Garcia-Nicolas, O., Zimmer, G., and Summerfield, A. (2023). NS1 and PA-X of H1N1/09 influenza virus act in a concerted manner to manipulate the innate immune response of porcine respiratory epithelial cells. *Front. Cell. Infect. Microbiol.* 13, 1222805. <https://doi.org/10.3389/fcimb.2023.1222805>.
7. Qin, T., Chen, Y., Huangfu, D., Yin, Y., Miao, X., Yin, Y., Chen, S., Peng, D., and Liu, X. (2022). PA-X protein of H1N1 subtype influenza virus disables the nasal mucosal dendritic cells for strengthening virulence. *Virulence* 13, 1928–1942. <https://doi.org/10.1080/21505594.2022.2139474>.
8. Gong, X.Q., Sun, Y.F., Ruan, B.Y., Liu, X.M., Wang, Q., Yang, H.M., Wang, S.Y., Zhang, P., Wang, X.H., Shan, T.L., et al. (2017). PA-X protein decreases replication and pathogenicity of swine influenza virus in cultured cells and mouse models. *Vet. Microbiol.* 205, 66–70. <https://doi.org/10.1016/j.vetmic.2017.05.004>.
9. Lee, J., Yu, H., Li, Y., Ma, J., Lang, Y., Duff, M., Henningson, J., Liu, Q., Li, Y., Nagy, A., et al. (2017). Impacts of different expressions of PA-X protein on 2009 pandemic H1N1 virus replication, pathogenicity and host immune responses. *Virology* 504, 25–35. <https://doi.org/10.1016/j.virol.2017.01.015>.
10. Ren, W., Zhang, C.H., Li, G., Liu, G., Shan, H., and Li, J. (2020). Two genetically similar H9N2 influenza viruses isolated from different species show similar virulence in minks but different virulence in mice. *Acta Virol.* 64, 67–77. https://doi.org/10.4149/av_2020_109.
11. Hussain, S., Turnbull, M.L., Wise, H.M., Jagger, B.W., Beard, P.M., Kovacicova, K., Taubenberger, J.K., Vervelde, L., Engelhardt, O.G., and Digard, P. (2019). Mutation of Influenza A Virus PA-X Decreases Pathogenicity in Chicken Embryos and Can Increase the Yield of Reassortant Candidate Vaccine Viruses. *Journal of Virology* 93. <https://doi.org/10.1128/JVI.01128-19>.
12. Li, M., Qi, W., Chang, Q., Chen, R., Zhen, D., Liao, M., Wen, J., and Deng, Y. (2021). Influenza A virus protein PA-X suppresses host Ankrd17-mediated immune responses. *Microbiol. Immunol.* 65, 48–59. <https://doi.org/10.1111/1348-0421.12863>.
13. Xu, G., Zhang, X., Sun, Y., Liu, Q., Sun, H., Xiong, X., Jiang, M., He, Q., Wang, Y., Pu, J., et al. (2016). Truncation of C-terminal 20 amino acids in PA-X contributes to adaptation of swine influenza virus in pigs. *Sci. Rep.* 6, 21845. <https://doi.org/10.1038/srep21845>.
14. Khapersky, D.A., and McCormick, C. (2015). Timing Is Everything: Coordinated Control of Host Shutoff by Influenza A Virus NS1 and PA-X Proteins. *J. Virol.* 89, 6528–6531. <https://doi.org/10.1128/jvi.00386-15>.
15. Oishi, K., Yamayoshi, S., and Kawaoka, Y. (2015). Mapping of a Region of the PA-X Protein of Influenza A Virus That Is Important for Its Shutoff Activity. *J. Virol.* 89, 8661–8665. <https://doi.org/10.1128/jvi.01132-15>.
16. Khapersky, D.A., Emara, M.M., Johnston, B.P., Anderson, P., Hatchette, T.F., and McCormick, C. (2014). Influenza A Virus Host Shutoff Disables Antiviral Stress-Induced Translation Arrest. *PLoS Pathog.* 10, e1004217. <https://doi.org/10.1371/journal.ppat.1004217>.
17. Hayashi, T., Chaimayo, C., McGuinness, J., and Takimoto, T. (2016). Critical Role of the PA-X C-Terminal Domain of Influenza A Virus in Its Subcellular Localization and Shutoff Activity. *J. Virol.* 90, 7131–7141. <https://doi.org/10.1128/JVI.00954-16>.
18. Oishi, K., Yamayoshi, S., Kozuka-Hata, H., Oyama, M., and Kawaoka, Y. (2018). N-Terminal Acetylation by NatB Is Required for the Shutoff Activity of Influenza A Virus PA-X. *Cell Rep.* 24, 851–860. <https://doi.org/10.1016/j.celrep.2018.06.078>.
19. Hu, Y., Chen, X., Ling, Y., Zhou, K., Han, M., Wang, X., Yue, M., and Li, Y. (2023). Influenza A virus inhibits TET2 expression by endoribonuclease PA-X to attenuate type I interferon signaling and promote viral replication. *PLoS Pathog.* 19, e1011550. <https://doi.org/10.1371/journal.ppat.1011550>.
20. Oishi, K., Yamayoshi, S., and Kawaoka, Y. (2019). Identification of amino acid residues in influenza A virus PA-X that contribute to enhanced shutoff activity. *Front. Microbiol.* 10, 432. <https://doi.org/10.3389/fmicb.2019.00432>.
21. Oishi, K., Yamayoshi, S., and Kawaoka, Y. (2018). Identification of novel amino acid residues of influenza virus PA-X that are important for PA-X shutoff activity by using yeast. *Virology* 516, 71–75. <https://doi.org/10.1016/j.virol.2018.01.004>.
22. Chiem, K., López-García, D., Ortego, J., Martínez-Sobrido, L., Dediego, M.L., and Nogales, A. (2022). Identification of Amino Acid Residues Required for Inhibition of Host Gene Expression by Influenza Virus A/Viet Nam/1203/2004 H5N1 PA-X. *J. Virol.* 96. <https://doi.org/10.1128/JVI.00408-21>.
23. Sparrer, K.M.J., and Kirchhoff, F. (2023). Cap Snatch Prevention: A Novel Approach to Tackle Influenza Viruses. *Signal Transduction and Targeted Therapy* 8. <https://doi.org/10.1038/s41392-023-01474-9>.
24. Noshi, T., Kitano, M., Taniguchi, K., Yamamoto, A., Omoto, S., Baba, K., Hashimoto, T., Ishida, K., Kushima, Y., Hattori, K., et al. (2018). In vitro characterization of baloxavir acid, a first-in-class cap-dependent endonuclease inhibitor of the influenza virus polymerase PA subunit. *Antiviral Res.* 160, 109–117. <https://doi.org/10.1016/j.antiviral.2018.10.008>.
25. Case, J.R., and Khapersky, D.A. (2026). Impaired host shutoff is a fitness cost associated with baloxavir marboxil resistance mutations in influenza A virus PA/PA-X nuclease domain. *PLoS Pathog.* 22, e1013550. <https://doi.org/10.1371/journal.ppat.1013550>.
26. Shi, M., Jagger, B.W., Wise, H.M., Digard, P., Holmes, E.C., and Taubenberger, J.K. (2012). Evolutionary Conservation of the PA-X Open Reading Frame in Segment 3 of Influenza A Virus. *J. Virol.* 86, 12411–12413. <https://doi.org/10.1128/jvi.01677-12>.
27. Sato, K., Akiyama, M., and Sakakibara, Y. (2021). RNA secondary structure prediction using deep learning with thermodynamic integration. *Nat. Commun.* 12, 941. <https://doi.org/10.1038/s41467-021-21194-4>.
28. Hatta, M., Asano, Y., Masunaga, K., Ito, T., Okazaki, K., Toyoda, T., Kawaoka, Y., Ishihama, A., and Kida, H. (2000). Epitope mapping of the influenza A virus RNA polymerase PA using monoclonal antibodies. *Arch. Virol.* 145. <https://doi.org/10.1007/s007050050687>.
29. Dias, A., Bouvier, D., Crépin, T., McCarthy, A.A., Hart, D.J., Baudin, F., Cusack, S., and Ruigrok, R.W.H. (2009). The cap-snatching endonuclease of influenza virus polymerase resides in the PA subunit. *Nature* 458, 914–918. <https://doi.org/10.1038/nature07745>.
30. Yuan, S., Chu, H., Singh, K., Zhao, H., Zhang, K., Kao, R.Y.T., Chow, B.K.C., Zhou, J., and Zheng, B.J. (2016). A novel small-molecule inhibitor of influenza A virus acts by suppressing PA endonuclease activity of the viral polymerase. *Sci. Rep.* 6, 22880. <https://doi.org/10.1038/srep22880>.
31. Baughman, B.M., Jake Slavish, P., Dubois, R.M., Boyd, V.A., White, S.W., and Webb, T.R. (2012). Identification of influenza endonuclease inhibitors using a novel fluorescence polarization assay. *ACS Chem. Biol.* 7, 526–534. <https://doi.org/10.1021/cb200439z>.
32. Kuang, W., Tian, Z., Zhang, G., Wu, F., Li, J., Tang, J., Zhang, H., Zhuo, X., Hu, Z., Wang, M., et al. (2026). Structure and function of the nairovirus

- cap-snatching endonuclease. *Nucleic Acids Res.* 54, gkaf1515. <https://doi.org/10.1093/nar/gkaf1515>.
33. Bousset, L., Mary, C., Brooks, M.A., Scherrer, A., Strub, K., and Cusack, S. (2014). Crystal structure of a signal recognition particle Alu domain in the elongation arrest conformation. *RNA* 20, 1955–1962. <https://doi.org/10.1261/ma.047209.114>.
34. Killip, M.J., Fodor, E., and Randall, R.E. (2015). Influenza virus activation of the interferon system. *Virus Res.* 209, 11–22. <https://doi.org/10.1016/j.virusres.2015.02.003>.
35. Gaglia, M.M., Covarrubias, S., Wong, W., and Glaunsinger, B.A. (2012). A Common Strategy for Host RNA Degradation by Divergent Viruses. *J. Virol.* 86, 9527–9530. <https://doi.org/10.1128/jvi.01230-12>.
36. He, T., Wang, M., Cheng, A., Yang, Q., Wu, Y., Jia, R., Liu, M., Zhu, D., Chen, S., Zhang, S., et al. (2020). Host Shutoff Activity of VHS and SOX-like Proteins: Role in Viral Survival and Immune Evasion. *Virology journal* 17. <https://doi.org/10.1186/s12985-020-01336-8>.
37. Abernathy, E., and Glaunsinger, B. (2015). Emerging Roles for RNA Degradation in Viral Replication and Antiviral Defense. *Virology* 479. <https://doi.org/10.1016/j.virol.2015.02.007>.
38. Richner, J.M., Clyde, K., Pezda, A.C., Cheng, B.Y.H., Wang, T., Kumar, G.R., Covarrubias, S., Coscoy, L., and Glaunsinger, B. (2011). Global mRNA degradation during lytic gammaherpesvirus infection contributes to establishment of viral latency. *PLoS Pathog.* 7, e1002150. <https://doi.org/10.1371/journal.ppat.1002150>.
39. Abernathy, E., Clyde, K., Yeasmin, R., Krug, L.T., Burlingame, A., Coscoy, L., and Glaunsinger, B. (2014). Gammaherpesviral Gene Expression and Virion Composition Are Broadly Controlled by Accelerated mRNA Degradation. *PLoS Pathog.* 10, e1003882. <https://doi.org/10.1371/journal.ppat.1003882>.
40. Horst, D., Burmeister, W.P., Boer, I.G.J., van Leeuwen, D., Buisson, M., Goralenya, A.E., Wiertz, E.J.H.J., and Rensing, M.E. (2012). The “Bridge” in the Epstein-Barr Virus Alkaline Exonuclease Protein BGLF5 Contributes to Shutoff Activity during Productive Infection. *J. Virol.* 86, 9175–9187. <https://doi.org/10.1128/jvi.00309-12>.
41. Wandzik, J.M., Kouba, T., and Cusack, S. (2021). Structure and function of influenza polymerase. *Cold Spring Harb. Perspect. Med.* 11, a038372. <https://doi.org/10.1101/cshperspect.a038372>.
42. Stevaert, A., and Naesens, L. (2016). The Influenza Virus Polymerase Complex: An Update on its Structure, Functions, and Significance for Antiviral Drug Design. *Medicinal research reviews* 36. <https://doi.org/10.1002/med.21401>.
43. Zhu, Z., Fodor, E., and Keown, J.R. (2023). A Structural Understanding of Influenza Virus Genome Replication. *Trends in microbiology* 31. <https://doi.org/10.1016/j.tim.2022.09.015>.
44. Noda, T., Murakami, S., Nakatsu, S., Imai, H., Muramoto, Y., Shindo, K., Sagara, H., and Kawaoka, Y. (2018). Importance of the 1+7 configuration of ribonucleoprotein complexes for influenza A virus genome packaging. *Nat. Commun.* 9, 54. <https://doi.org/10.1038/s41467-017-02517-w>.
45. Nakano, M., Sugita, Y., Koda, N., Miyamoto, S., Muramoto, Y., Wolf, M., and Noda, T. (2021). Ultrastructure of influenza virus ribonucleoprotein complexes during viral RNA synthesis. *Commun. Biol.* 4, 858. <https://doi.org/10.1038/s42003-021-02388-4>.
46. Ozawa, M., Fujii, K., Muramoto, Y., Yamada, S., Yamayoshi, S., Takada, A., Goto, H., Horimoto, T., and Kawaoka, Y. (2007). Contributions of Two Nuclear Localization Signals of Influenza A Virus Nucleoprotein to Viral Replication. *J. Virol.* 81, 30–41. <https://doi.org/10.1128/jvi.01434-06>.
47. Boulo, S., Akarsu, H., Ruigrok, R.W.H., and Baudin, F. (2007). Nuclear Traffic of Influenza Virus Proteins and Ribonucleoprotein Complexes. *Virus research* 124. <https://doi.org/10.1016/j.virusres.2006.09.013>.
48. Bayer, T.S., Booth, L.N., Knudsen, S.M., and Ellington, A.D. (2005). Arginine-rich motifs present multiple interfaces for specific binding by RNA. *RNA* 11, 1848–1857. <https://doi.org/10.1261/rna.2167605>.
49. Daly, R.R., Feng, C.Y., Hesser, C.R., Myasnikov, I., and Gaglia, M.M. (2026). N-terminal acetylation controls multiple functional aspects of the influenza A virus ribonuclease PA-X. *J. Virol.* 100. <https://doi.org/10.1128/jvi.01999-25>.
50. Schindelin, J., Arganda-Carreras, I., Frise, E., Kaynig, V., Longair, M., Pietzsch, T., Preibisch, S., Rueden, C., Saalfeld, S., Schmid, B., et al. (2012). Fiji: an open-source platform for biological-image analysis. *Nat. Methods* 9, 676–682. <https://doi.org/10.1038/nmeth.2019>.
51. Chaimayo, C., Dunagan, M., Hayashi, T., Santoso, N., and Takimoto, T. (2018). Specificity and functional interplay between influenza virus PA-X and NS1 shutoff activity. *PLoS Pathog.* 14, e1007465. <https://doi.org/10.1371/journal.ppat.1007465>.
52. Omoto, S., Speranzini, V., Hashimoto, T., Noshi, T., Yamaguchi, H., Kawai, M., Kawaguchi, K., Uehara, T., Shishido, T., Naito, A., and Cusack, S. (2018). Characterization of influenza virus variants induced by treatment with the endonuclease inhibitor baloxavir marboxil. *Sci. Rep.* 8, 9633. <https://doi.org/10.1038/s41598-018-27890-4>.
53. Kwong, A.D., Kruper, J.A., and Frenkel, N. (1988). Herpes Simplex Virus Virion Host Shutoff Function. *J. Virol.* 62, 912–921.
54. Jansen, A., Bruse, N., Waalders, N., Gerretsen, J., Rijbroek, D., Pickkers, P., and Kox, M. (2023). Ex vivo and in vitro Monocyte Responses Do Not Reflect in vivo Immune Responses and Tolerance. *J. Innate Immun.* 15, 174–187. <https://doi.org/10.1159/000525572>.
55. Chen, Y., and Nielsen, J. (2021). In vitro turnover numbers do not reflect in vivo activities of yeast enzymes. *Proc. Natl. Acad. Sci. USA* 118, 2108391118. <https://doi.org/10.1073/pnas.2108391118/-DCSupplemental>.
56. Danielsson, J., and Oliveberg, M. (2017). Comparing protein behaviour in vitro and in vivo, what does the data really tell us? *Curr. Opin. Struct. Biol.* 42, 129–135. <https://doi.org/10.1016/j.sbi.2017.01.002>.
57. García-Contreras, R., Vos, P., Westerhoff, H.V., and Booger, F.C. (2012). Why in vivo may not equal in vitro - New effectors revealed by measurement of enzymatic activities under the same in vivo-like assay conditions. *FEBS J.* 279, 4145–4159. <https://doi.org/10.1111/febs.12007>.
58. Todd, B., Tchesnokov, E.P., and Götte, M. (2021). The active form of the influenza cap-snatching endonuclease inhibitor baloxavir marboxil is a tight binding inhibitor. *J. Biol. Chem.* 296, 100486. <https://doi.org/10.1016/j.jbc.2021.100486>.
59. Gaucherand, L., Iyer, A., Gilabert, I., Rycroft, C.H., and Gaglia, M.M. (2023). Cut site preference allows influenza A virus PA-X to discriminate between host and viral mRNAs. *Nat. Microbiol.* 8, 1304–1317. <https://doi.org/10.1038/s41564-023-01409-8>.
60. Oishi, K., Blanco-Melo, D., Kurland, A.P., Johnson, J.R., and tenOever, B.R. (2023). Archaeal Kink-Turn Binding Protein Mediates Inhibition of Orthomyxovirus Splicing Biology. *J. Virol.* 97, e0181322. <https://doi.org/10.1128/jvi.01813-22>.

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Prime for Reverse transcription, see Table 3	This paper	N/A
Prime for qPCR, see Table 3	This paper	N/A
Prime for conventional PCR, see Table 3	This paper	N/A
Recombinant DNA		
M13 mp18 single-stranded DNA	Takara	Cat#3518
Software and algorithms		
Fiji	Schindelin et al. ⁵⁰	https://imagej.net/software/fiji/downloads
GraphPad	GraphPad Software	https://www.graphpad.com
MXfold2	Sato et al. ²⁷	https://ws.sato-lab.org/mxfold2/
Octet Data Analysis Software v10.0.3.1	ForteBio	https://www.sartorius.com/en/products/biolayer-interferometry/octet-systems-software

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cells

Human embryonic kidney 293T (HEK293T) cells (ATCC: CRL-3216) and Huh-7 cells (JCRB cell bank: LCRB0403) were maintained in DMEM (Sigma) supplemented with 10% FCS and penicillin-streptomycin. A549 cells (ATCC: CCL-185) were maintained in Ham's F-12K (Wako) supplemented with 10% FCS and penicillin-streptomycin. Madin-Darby canine kidney (MDCK) cells were maintained in Eagle's minimal essential medium (MEM) containing 5% newborn calf serum (NCS). Expi293F cells (ThermoFisher) used for recombinant protein expression were maintained in Expi293 Expression Medium at 37°C and 8% CO₂ on a rotator set at 125 rpm. These cell lines were obtained directly from the indicated approved repositories (ATCC, JCRB cell bank, or ThermoFisher), and further authentication was not performed. All cell lines were routinely tested and confirmed negative for mycoplasma contamination.

METHOD DETAILS

Shutoff assay

293T cells were co-transfected with a reporter plasmid expressing firefly luciferase or GFP together with either an empty vector or a C-terminally FLAG-tagged PA-X expression plasmid. Cells were treated with DMSO or baloxavir acid (BXA; Shionogi & Co., LTD.) at the time of transfection. At 48 h post-transfection, GFP signals were observed by using a fluorescence microscope (BZ-X710). Luciferase activity was measured using the ONE-Glo Luciferase Assay System (Promega) and GloMax Discovery System (Promega) at 48 h post-transfection. The shutoff activity of PA-X was calculated by dividing the luminescence value of firefly luciferase co-transfected with an empty plasmid by that co-transfected with PA-X-expressing plasmids. Shutoff activities are presented as mean values $\pm$ SD [$n = 9$ ($n = 3$ technical replicates, $n = 3$ biological triplicates)]. To analyze the expression of PA-X and GFP, transfected cells were lysed in 2 $\times$ SDS sample buffer (Invitrogen) at 48 h post-transfection. Cell lysates were sonicated, incubated for 10 min at 95°C, and then loaded onto a 5%–20% e-PAGE (ATTO). Separated proteins were transferred onto Immobilon-P PVDF membrane (ThermoFisher) and detected by using anti-DYKDDDDK (FLAG) tag antibody (clone 1E6; Fujifilm), anti-GFP polyclonal antibody (MBL life science), or anti-GAPDH antibody (clone 10494-1-AP; Proteintech), followed by Goat anti-mouse IgG-HRP (LGC clinical diagnosis) or Goat anti-rabbit IgG-HRP (Vector laboratories).

Purification of PA-X proteins

Expi293F cells were transfected with a plasmid encoding PA-X derived from the A/Puerto Rico/8/34 (H1N1) strain (GenBank accession no. EF467820.1) with a C terminus TEV-6 $\times$ His-FLAG by using the ExpiFectamine 293 transfection kit (Gibco). Eighteen hours post-transfection, BXA was added to the culture medium at a final concentration of 1 μ M. To completely remove contaminating host nucleases, proteins were purified by two-steps immunoprecipitation using both His and FLAG tags. At 48 h post-transfection, cells were harvested and lysed in lysis buffer [50 mM Tris-HCl (pH 7.5), 150 mM NaCl, 0.5% Nonidet P-40, and protease inhibitor Complete Mini cocktail (Roche)] for 2 h at 4°C. After centrifugation to remove cellular debris, the supernatants were incubated with TALON (Cobalt) Metal Affinity resin (Takara) at 4°C. After an incubation for 2 h, samples were loaded onto Pierce centrifuge columns (ThermoFisher). The resins were washed twice with wash buffer [50 mM Tris-HCl (pH 7.5), 500 mM NaCl and 10 mM imidazole] and bound protein was eluted with elution buffer [50 mM Tris-HCl (pH 7.5), 500 mM NaCl and 400 mM imidazole]. The eluate was concentrated using a Vivaspin 20 centrifugal concentrator (10 kDa MWCO; Cytiva). The concentrated elution was diluted with PBS and incubated with anti-FLAG M2 magnetic beads (Sigma) overnight at 4°C. The magnetic beads were washed three times

with wash buffer [50 mM Tris-HCl (pH 7.5) and 150 mM NaCl]. The protein was eluted by incubation with FLAG elution buffer [50 mM Tris-HCl (pH 7.5), 150 mM NaCl, and 0.5 mg/mL FLAG peptide (Millipore)] for 2 h at 4°C. The final elution was concentrated using an Amicon Ultra centrifugal Filter (10 kDa MWCO; Millipore). A portion of the purified protein was mixed in 2×SDS sample buffer and denatured at 95°C for 10 min. Samples were then subjected to SDS-PAGE (5%–20% e-PAGE; ATTO), followed by CBB staining (Protein Safe Stain; Takara). Protein concentrations were determined using a Nanodrop and validated by CBB staining using a known concentration of BSA (NEB) as a standard.

In vitro endonuclease assay

Nucleic acid substrates were incubated with either purified protein, an equal amount of BSA, or one-tenth amounts of RNase A at 37°C for 2 h. The standard reaction buffer consisted of 20 mM of Tris-HCl (pH 8.0), 10 mM NaCl, 1 mM β-mercaptoethanol, and 1 mM of various metal salts. To evaluate endonuclease inhibitors, BXA, DPBA (Fujifilm), or DMSO was added to the reaction mixture. ssDNA (M13 mp18 single stranded DNA) (Takara) and dsRNA (Jena Bioscience) were purchased from commercial sources. ssRNAs (Table 2) were chemically synthesized (Merck). RNAs with or without a cap structure and/or a poly A tail were synthesized by *in vitro* transcription. DNA fragments encoding the GFP ORF with or without a poly A tail were amplified by PCR and used as templates for *in vitro* transcription. Capped or uncapped RNA was synthesized using the IVTpro mRNA Synthesis System (Takara) and CleanCap Reagent AG (Fujifilm). For the analysis of ssDNA and dsRNA, reactions were quenched with loading dye containing EDTA, and products were resolved on 1.0% agarose gel followed by GelRed staining. For ssRNA analysis, reactions were terminated by adding RNA gel loading dye (ThermoFisher). Samples were denatured at 70°C for 10 min and then immediately chilled on ice. Samples were loaded onto 20% polyacrylamide and 8 M urea gel. Gels were stained with EzPreStain DNA & RNA (ATTO). Band intensities were quantified by using Fiji software.

Reaction of PA-X with vRNAs and vRNPs

MDCK cells were infected with influenza A virus (A/Puerto Rico/08/34) at a low multiplicity of infection (MOI = 0.01–0.001).⁶⁰ After a 48 h post-infection, the culture supernatant was clarified by centrifugation. Virions were then concentrated by ultracentrifugation through a 25% sucrose cushion at 28,000 rpm for 2 h at 4°C. The resulting pellets were suspended in phosphate-buffered saline (PBS). For the preparation of naked viral RNA (vRNA), total RNA was isolated from the resuspended virions using Trizol-LS Reagent (ThermoFisher). To isolate vRNP complexes, the resuspended virions were incubated in a disruption buffer [50 mM Tris-HCl pH 8.0, 100 mM KCl, 5 mM MgCl₂, 1 mM dithiothreitol (DTT), 2% lyssolecithin, 2% Triton X-100, 5% glycerol, and RNase OUT (Invitrogen)] for 1 h at 30°C, as previously described.^{44,45} The samples were then subjected to 30%–60% glycerol density gradient ultracentrifugation at 45,000 rpm for 3 h at 4°C. Following centrifugation, 60 μL of each fraction was collected and subjected to SDS-PAGE followed by CBB staining. Fractions 6 and 7 were pooled as purified vRNPs based on the consistent co-elution of the polymerase complex and NP protein, as confirmed by SDS-PAGE. For the endonuclease assay, naked vRNAs or isolated vRNP complexes were incubated with purified PA-X, an equal amount of BSA, or one-tenth amounts of RNase A at 37°C for 2 h. The reaction buffer consisted of 20 mM of Tris-HCl pH 8.0, 10 mM NaCl, 1 mM β-mercaptoethanol, 1 mM MnCl₂ and either 1 μM BXA or DMSO. After incubation, RNA was isolated from each reaction and viral RNA was reverse-transcribed into cDNA using specific primer Au-12 (Table 3). The copy number of NP vRNA was calculated by TaqMan-based qPCR and normalized using an *in vitro* transcribed standard NP vRNA standard. HA and NA vRNA segments were amplified by PCR using the specific primers listed in Table 3.

Table 3. Primers sequences used in this study

Primer/Probe	Sequence
Au-12	AGCAAAGCAGG
NP probe	FAM-GTCCTGCCTGCCTGCCTGTG-BHQ-1
NP-F	TCCCAGGATGTGCTCTCTGA
NP-R	GCCATCCACACCAAGTTGACT
HA-F	AGCAAAGCAGGGGAAAATAAAAGCAAC
HA-R	AGTAGAAACAAAGGGTGTCTTCTCATG
NA-F	AGCAAAGCAGGAGTTCAAAATGAATC
NA-R	TTATATGGTCTCGTATTAGTAGAAACAAGG

Immunofluorescent assay

Cells were transfected with plasmids encoding FLAG-tagged PA-X, FLAG-tagged PA-X 6E, or His-tagged PA-X 6E. At 48 h post-transfection, cells were fixed with 4% paraformaldehyde and permeabilized with 0.2% Triton X-100. For protein detection, cells were incubated with an anti-DYKDDDDK (FLAG) mouse monoclonal antibody (clone 1E6; Sigma) or an anti-His rabbit monoclonal antibody (clone D3110; Cell Signaling Technology), followed by Alexa Fluor 488-conjugated goat anti-mouse IgG (Invitrogen), or

by Alexa Fluor 555-conjugated goat anti-rabbit IgG (Invitrogen), respectively. Nuclei were counterstained with Hoechst 33342 (Invitrogen). Fluorescence images were acquired using an FLUOVIEW FV3000 confocal microscope (Olympus).

Electrophoretic mobility shift assay

Synthetic RNAs (Merck) were 5' end-labeled with [γ ³²P] ATP using T4 polynucleotide kinase (NEB). Labeled RNAs were purified from unincorporated [γ ³²P] ATP using the RNA Clean and Concentrator kit (Zymo Research). For the binding reaction, 15 nM of labeled RNA was incubated with various concentrations of purified protein in binding buffer [10 mM Tris-HCl pH 7.5, 40 mM NaCl, and 1 mM DTT] for 1 h at room temperature. Protein-RNA complexes were resolved by native gel electrophoresis on 6% polyacrylamide gels. Signals were visualized by phosphor screen autoradiography using a Typhoon scanner (GE Healthcare). The intensities of the free RNA and shifted bands were quantified using Fiji software (ImageJ).

Biolayer interferometry analysis

The affinities of RNA molecules for the PA-X and PA-X 6E proteins were evaluated using an Octet RED96e instrument (FortéBio, Fremont, CA). Biotin tagged-synthesized RNA served as ligands, while recombinant PA-X proteins (prepared as described above) were used as analytes. Assays were conducted at 30°C with shaking at 1,000 rpm in Octet Kinetics Buffer (FortéBio) using streptavidin (SA) biosensors (FortéBio). RNA (500 nM) was loaded onto surfaces of pre-hydrated biosensors for 10 min, and after a 1-min baseline equilibration step, association with PA-X protein was permitted for 5 min, followed by dissociation for 10 min in kinetics buffer. Kinetic assays were performed at concentrations ranging from 62.5 nM to 7.8125 nM for PA-X and from 500 nM to 7.8125 nM for PA-X 6E (2-fold serial dilutions). Data were corrected by double reference subtraction, and a 1:1 binding model with global fit was used to determine affinity constants (KD, kon, and kdis) using Octet Data Analysis Software v10.0.3.1 (FortéBio).

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical details of experiments, including the specific statistical tests used, exact value of n, and what n represents, are described in the respective figure legends. Data are presented as mean $\pm$ SD. Significance was determined using one-way ANOVA followed by Tukey's or Dunnett's multiple comparison tests, or a two-tailed unpaired Student's *t* test, as appropriate. All analyses were performed using Prism software (GraphPad). A value of *p* < 0.05 was considered statistically significant.